

CHAPTER 2 · CONCEPT 4

Microbes in Our Lives

Microbes make curd and bread, enrich soil, clean waste — and a few cause spoilage and disease.

Friendly microbes at work

Most microbes are **helpful**. They touch nearly every part of our lives.

Microbe	What it does
Yeast (fungus)	Respires and releases carbon dioxide , making dough soft and fluffy (bread, cakes); also a little alcohol → smell. Needs warmth & sugar.
Lactobacillus (bacterium)	Turns milk into curd ; produces lactic acid → sour taste. Works in warm conditions.
Rhizobium (bacterium)	Lives in root nodules of legumes (peas, beans, lentils); traps nitrogen from air to enrich soil.
Bacteria & fungi in soil	Decompose dead plants/animals into nutrient-rich manure ; recycle nutrients.
Microalgae (e.g. Spirulina)	Release oxygen (over half the Earth's O ₂); food/health supplements; clean water; biofuel.

Idli, dosa, bhatura: made by fermentation — Lactobacillus and yeast at work. Biogas (methane + CO₂) is made by bacteria breaking down waste without oxygen.

A few harmful microbes

Some microbes **spoil food** (mould on fruit) and some cause **disease** in plants, animals and humans. Spoilage is prevented by drying, salting, adding sugar (pickles, murabba) or refrigeration — these stop microbes growing.

Why pickles don't spoil: high salt/sugar concentration does not let microbes grow on them — salt and sugar act as preservatives.

Where students lose marks

Trap 1 — Yeast gives CO₂, Lactobacillus gives lactic acid. Yeast makes dough rise (gas); Lactobacillus makes curd sour (acid). Don't mix them up.

Trap 2 — Rhizobium fixes nitrogen, in legume root nodules. Not in cereals; that's why farmers rotate legumes with other crops.

Trap 3 — Decomposition vs manure. Microbes break complex dead matter into simpler nutrients (decomposition); the dark, nutrient-rich result is manure.

Preservation: salt/sugar/cold stop microbes growing — that is why curd forms in warm milk but not in the fridge.

Model answers — write them like this

Example A. Why does dough with yeast rise and smell slightly different?

Step 1: Yeast respires, breaking down sugar and releasing carbon dioxide gas.

Step 2: CO₂ bubbles make the dough fluffy; a little alcohol is also formed.

∴ CO₂ makes it rise; the alcohol gives the smell.

Example B. Why do farmers grow legumes (peas/beans) in rotation with other crops?

Step 1: Legume root nodules contain Rhizobium bacteria.

Step 2: Rhizobium traps nitrogen from air and enriches the soil naturally.

∴ Rotation adds nitrogen to soil, so less chemical fertiliser is needed.

Example C. Why does milk turn to curd in a warm place but not in the fridge?

Step 1: Lactobacillus in curd ferments milk and forms lactic acid.

Step 2: These bacteria grow well in warmth, but cold slows them down.

∴ Warmth lets Lactobacillus act, so curd sets; cold prevents it.

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